Resonant acousto-optic frequency shifting in a thin-film lithium niobate photonic molecule: design, tolerances, and a measurement plan

Xinlei Chen[1] and Yuanhua Li[1]

[1] *Shanghai University of Electric Power, Shanghai, China*

(Dated: October 14, 2026)

We present a design and measurement plan for a thin-film lithium niobate (TFLN) frequency shifter in which a suspended interdigital-transducer (IDT) acoustic resonator drives one ring of a dual-ring photonic molecule in a forward, single-bus geometry at a fixed shift of 3.33 GHz, intended for fabrication. All device numbers below are simulation predictions. Against an electro-optic (EO) drive of the same optics, the Shao-calibrated IDT needs ≈6.7 μW while a matched EO feed needs 1.2–13 μW: we do *not* claim an acousto-optic (AO) power advantage; the AO RF bandwidth is ∼1 MHz. Versus the measured traveling-wave AO shifter of Shao *et al.* (3.5% on-chip absolute efficiency at 1 W, 70 MHz bandwidth), we compare absolute efficiency only and state in the same sentence that our result is simulated at ∼1 MHz bandwidth. With a self-labeled thermo-optic Mach–Zehnder (MZI) ring–ring coupler that tunes the supermode splitting 2μ (heater on; $P_\pi = 44.4$ mW as a Maeder geometric assumption), a 1000-sample Monte Carlo under conservative extra loss ($\kappa_i \times 2$) and 10% IDT efficiency predicts a median $\eta_{\text{shift}} = 96.4\%$ (10th–90th percentile 91.7–98.3%); the heater-off fixed-gap case (median 94.5%; 74.2% of samples at $\eta_{\text{shift}} \geq 90\%$) is retained only as a comparison. Insertion loss is reported in two columns: Shao 2019 suspended-racetrack Q_i as measured (heater-on median 2.48 dB) and assumed extra $\kappa_i \times 2$ (median 4.72 dB). A partial-coverage factor ξ multiplies the extrapolated $\delta\omega/2\pi = 21.3$ GHz/V ($\xi = 1$); at $\eta_{\text{IDT}} = 10\%$ and $Q_m = 1570$, source power is 478/172/43/11 μW for $\xi = 0.3/0.5/1/2$. For stage-1 spectral tests, unfiltered $1 - V \approx 1 - \eta_{\text{shift}}$ (ideal $\approx 1.7 \times 10^{-2}$). Thermal phonons give a drive jitter $n_{\text{th}}/N_{\text{coh}} \approx 1.07 \times 10^{-8}$ and an upper bound $n_{\text{th}} C_0 \approx 8.0 \times 10^{-6}$ for spontaneous scattering on the same acoustic mode ($g_0/2\pi = 1.1$ kHz), below the unfiltered sideband error.

I. INTRODUCTION

Photonic qubits encoded in discrete frequency bins need low-loss, coherent devices that move light—and, in quantum networks, single photons—between bins [1–3]. Single-photon frequency shifting is a motivating application of this design, not a claim of a completed experiment. TFLN combines strong Pockels, piezoelectric, and photoelastic responses with low-loss resonators [4, 5], and has been used for single-photon spectral shearing over ±641 GHz [6], an integrated quantum frequency processor [7], and coupled-cavity frequency beam splitters [8]. For single-tone shifting, the benchmark is the EO coupled-ring shifter of Hu *et al.* [9]: a microwave drive couples the symmetric (S) and antisymmetric (AS) supermodes of a photonic molecule [10] and, at a generalized critical-coupling condition, transfers light unidirectionally from one supermode to the other. The measured device shifts 98.7% of the output light by 28.2 GHz with 0.45 dB insertion loss (IL), and 99.1% (up) and 99.4% (down) at 12.5 GHz, using 96 mW of microwave power [9]. A different EO approach, serrodyne driving of cascaded traveling-wave TFLN phase modulators, reports a shifting efficiency above 98.8% with $V_\pi \approx 2.3$ V [11].

Acoustic frequency shifting on TFLN is much less mature. The best published TFLN acousto-optic frequency shifter, a traveling-wave Bragg-deflection device at ∼3 GHz, reaches an on-chip efficiency of 3.5% at 30 dBm (1 W) applied to the IDT pads, with >30 dB carrier suppression and a 70-MHz microwave bandwidth [12]. Resonant acoustics improves the per-volt modulation strongly: a suspended 100-μm acoustic resonator at 3.33 GHz gives $V_\pi = 4.6$ V ($V_\pi L = 0.046$ V cm) on one arm of a Mach–Zehnder interferometer (MZI) [13]. Acoustic waves have long been used to modulate integrated resonators [14] and to control photonic molecules dynamically [15].

Most relevant here, Zhu *et al.* [16] have already driven a lithium niobate (LN-on-sapphire) coupled-microring photonic molecule acoustically. In their device an IDT-launched 8.54-GHz acoustic wave forms a backward-Brillouin dynamic Bragg mirror that couples forward- and backward-propagating light, reaching strong supermode coupling at milliwatt-level acoustic power (cooperativity 2.46 mW^{-1}) and Bragg reflectivities up to 24%. Acoustic driving of an LN photonic molecule is therefore not new. The distinction of the present design is its operating point: a *forward* (co-directional), *single-bus* frequency shifter that sends ≈98% of the output into one sideband, analyzed in terms of shift efficiency, insertion loss, spurious orders, and the RF power at the source. We also considered an all-optical, IDT-free alternative in which backward stimulated Brillouin scattering in TFLN, pumped by an EO frequency comb, writes the phonon; in our (unpublished) analysis it is not viable for multi-bin frequency operations, whereas forward-Brillouin frequency-bin swaps have been proposed theoretically [17].[18]

In this paper we (i) define the device (Fig. 1) and calibrate its drive to measured data [Sec. II]; (ii) compute its operating point and compare its source power with an EO

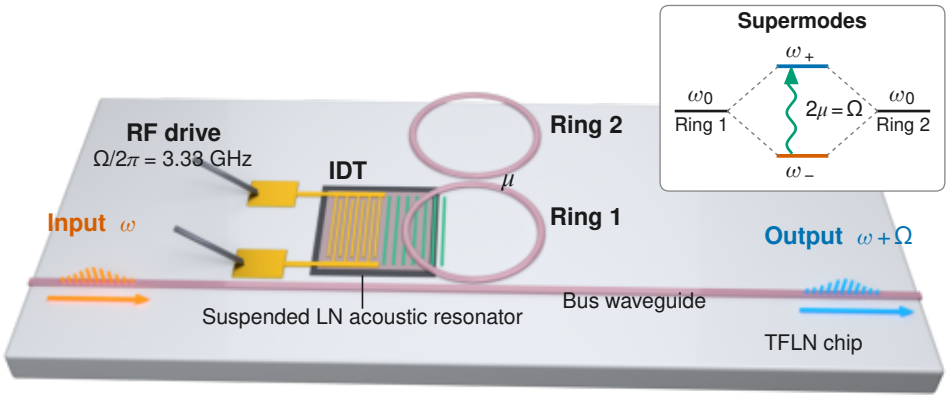


FIG. 1. Device concept. An RF-driven interdigital transducer (IDT; $\Omega/2\pi = 3.33$ GHz) excites a suspended LN acoustic resonator that covers only a segment of ring 1. Ring 2 is coupled only to ring 1 (coupling rate μ), and a single bus waveguide is coupled only to ring 1. Light enters in the forward direction at ω and leaves the same bus at the shifted frequency $\omega + \Omega$. Inset: the ring–ring coupling splits the two degenerate ring modes (ω_0) into supermodes $\omega_\pm$ separated by $2\mu = \Omega$, which the acoustic drive couples. Schematic, not to scale.

drive of identical optics, *including* impedance matching on both sides [Sec. III]; (iii) quantify tolerances and evaluate a heater-tunable coupler under realistic Monte Carlo conditions [Secs. IV and V]; (iv) give a five-step measurement plan with pass criteria for tape-out [Sec. VI]; and (v) report stage-2 HOM predictions and thermal-phonon bounds [Sec. VII]. Limitations are collected in Sec. VIII. All device numbers are simulation predictions until measured.

II. DEVICE AND MODEL

Structure.—The device (Fig. 1) is the coupled-ring photonic molecule of Ref. [9], with only ring 1 coupled to the bus waveguide. A 100-μm section of ring 1 passes through a suspended LN acoustic resonator driven by an IDT, with the geometry and calibration of Ref. [13]. The ring length is $L_{\rm ring} = 0.667$ mm, so the acoustic field covers 15% of the ring. This is the same coverage ratio, $\eta_{\rm cav} = 0.15$, as the suspended racetrack of Ref. [13]. We take $n_g = 2.3$. The ring–ring coupler is a *self-labeled* thermo-optic MZI with $\mu = \mu_{\rm gap}\sin(\Delta\phi/2)$, $\Delta\phi \in [0, \pi]$, designed with $\mu_{\rm gap} = \mu\sqrt{2}$ at the nominal gap so that a heater bias near $P_\pi/2$ can null $|2\mu - \Omega_m|$ after fabrication ($\Omega_m = 2\pi \times 3.33$ GHz). We do *not* take $\mu(\phi)$ from Ref. [10] (that work sets 2μ by the gap and uses electrodes for DC detuning). The coupler-arm $P_\pi = 44.4$ mW is a Maeder geometric assumption (NOI MZI heater arm [19]), not a calibration of this layout. Heater-induced differential and common-mode ring heating are folded into the existing δ_r and Δ model with assumed crosstalk after a coupler-isolation redesign (2 mK differential and 3 mK common at P_π; must be verified by heater calibration, not a measured result).

Equations of motion.—We use Eq. (S2) of Ref. [9], with ring 1 coupled to the bus at rate γ and both rings having intrinsic loss rate κ_i:

$$\begin{aligned}
\dot{a}_1 &= \Big[-i\big(\Delta + \delta\omega\cos\Omega_m t\big) - \tfrac{\gamma+\kappa_i}{2}\Big]a_1 - i\mu a_2 - \sqrt{\gamma}\,\alpha_{\rm in},\\
\dot{a}_2 &= \Big[-i\big(\Delta + s\,\delta\omega\cos\Omega_m t\big) - \tfrac{\kappa_i}{2}\Big]a_2 - i\mu a_1,\\
a_{\rm out} &= \alpha_{\rm in} + \sqrt{\gamma}\,a_1. \qquad (1)
\end{aligned}$$

For the acoustic (AO) drive only ring 1 is modulated ($s = 0$); for the EO reference we use Hu's push-pull drive ($s = -1$). The laser is tuned to the S supermode ($\Delta = \mu$). We integrate Eq. (1) with a fourth-order Runge–Kutta scheme (80 steps per modulation period, 40 ns total) without the rotating-wave approximation, and project $a_{\rm out}$, averaged over the final 20 ns, onto the frequencies $\omega_L + q\Omega_m$ ($|q| \le 3$). In the S/AS basis the single-ring term contains $\frac{\delta\omega}{2}\cos\Omega_m t\,(c_S^\dagger c_A + \text{h.c.})$ plus a common-mode phase modulation of both supermodes, so the supermode coupling is $\delta\omega/2$. As in Ref. [9], we define $\eta_{\rm shift} = P_{\rm shift}/P_{\rm out}$, the absolute efficiency $\eta_{\rm abs} = P_{\rm shift}/P_{\rm in}$, and IL $= -10\log_{10}(P_{\rm out}/P_{\rm in})$.

Drive calibration.—A peak voltage V from a 50-Ω source ($P_{\rm RF} = V^2/2R$, $R = 50\ \Omega$) produces the phase $\Delta\phi = \pi V L_a/(V_\pi L)$ over the covered length $L_a = 100\ \mu$m, which shifts the ring resonance by

$$\delta\omega = \Delta\phi\,\frac{c}{n_g L_{\rm ring}}. \qquad (2)$$

With $V_\pi L = 0.046$ V cm [13] this gives $\delta\omega/2\pi = 21.3$ GHz/V when the acoustic field covers the waveguide as in that MZI. We define a partial-coverage factor ξ so that the ring resonance shift is $\xi \times 21.3$ GHz/V; $\xi = 1$ is this straight-waveguide extrapolation at the acoustic Q used in the Shao 2019 modeling (typically 2000; measured up to 3600). Source power at fixed $\delta\omega$ then scales as $1/\xi^2$. We sweep $\xi = 0.3, 0.5, 1, 2$ and quote a conservative lab margin at $\xi = 0.5$ together with $\eta_{\rm IDT} = 10\%$ and

$Q_m = 1570$. Reference [13] extracts V_π from the measured S_{21} referenced to the power of a 50-Ω source, so the $\xi = 1$ calibration already contains the measured ≈50% electrical-to-acoustic coupling of that IDT (a −3 dB dip in S_{11}). We refer to this case as the "Shao IDT"; an ideally matched IDT would need half the power, and an IDT of efficiency η_{IDT} needs $0.5/\eta_{\mathrm{IDT}}$ times the Shao-IDT power. For the acoustic resonance we use $Q_m = 2000$ and $\gamma_e/\gamma = 0.15$, the values used to model the measured devices in Ref. [13] (measured acoustic Q up to 3600, Fig. 3(c) of Ref. [13]).

Optical parameters.—We take $\gamma/2\pi = 0.6$ GHz and the intrinsic loss implied by the suspended racetrack of Ref. [13]: from its Table S2, $\kappa/2\pi = 95$ MHz and $2\kappa_e/\kappa = 0.3$, so $\kappa_i/2\pi = 0.85 \times 95$ MHz $= 80.75$ MHz, i.e. $Q_i \approx 2.4 \times 10^6$ (at 193.4 THz). The main text of Ref. [13] quotes a loaded Q of 2.2×10^6 for the same linewidth.

EO reference with identical optics.—The EO drive uses the same optical parameters, push-pull modulation, and Hu's EO coefficient of ≈0.5 GHz/V referred to the voltage V_c on the electrode capacitor [9]. We model the electrode as Hu's lumped circuit, $C = 0.11$ pF, $L = 0.12$ nH, $R = 1.5$ Ω [9], whose reactance at 3.33 GHz is $X_C = 434$ Ω. We consider four feeds: (a) $V_c = V_0$, the convention of Ref. [9], with V_0 the source peak voltage; (b) an open-circuited capacitor, $V_c = 2V_0$; (c) a lumped LC match with inductor quality factor $Q_L = 30$ or 100, for which $P_{\mathrm{RF}} = V_c^2(R + X_C/Q_L)/(2X_C^2)$; and (d) an ideal inductor, $P_{\mathrm{RF}} = V_c^2 R/(2X_C^2)$. The $Q_L = 30$ match has an RF bandwidth of ≈122 MHz.

III. OPERATING POINT AND DRIVE POWER

Ideal operating point (upper bound).—Sections III and IV use nominal, idealized parameters (identical rings, no fabrication error, no extra loss, perfect temperature control); the realistic-condition results that we regard as the main prediction are given in Sec. V. At $\gamma/2\pi = 0.6$ GHz the acoustic drive reaches its maximum $\eta_{\mathrm{shift}} = 98.3\%$ at $\delta\omega/2\pi = 0.55$ GHz, i.e. a source peak voltage of 26 mV and $P_{\mathrm{RF}} = 6.7$ μW with the Shao IDT (Table I, Fig. 2). The absolute efficiency is 55% and the IL is 2.5 dB; both are set by the intrinsic loss rather than by the drive. The shifted line is the $q = +1$ order. Relative to it, the carrier is at −23.2 dB and the largest spurious order is $q = -1$ at −21.2 dB ($q = +2$ at −23.6 dB). These spurs come from the common-mode phase modulation of single-ring driving. The push-pull EO drive cancels this term and reaches $\eta_{\mathrm{shift}} = 99.6\%$, with the carrier at −26.0 dB and the $q = -1$ order at −30.6 dB [Fig. 2(b)], and the same 2.5 dB IL.

Comparison with EO driving.—Table II lists the source power at the respective optima for each feed. Without any matching on the EO side ($V_c = V_0$) the EO drive needs 3.1 mW, ≈460 times the 6.7 μW of the acoustic drive, and the measured device of Ref. [9] used 96 mW. These factors, however, compare a resonantly enhanced acoustic drive with a non-resonant electrode, and they do not survive a like-for-like comparison. With an LC match the EO power falls to 13 μW ($Q_L = 30$), 4.8 μW ($Q_L = 100$), or 1.2 μW (ideal inductor). Against the ideally matched IDT (3.4 μW), the two drives therefore differ only by a factor of 0.36–3.9, and *the EO drive is the more power-efficient one when the inductor loss is low.* Against the as-measured Shao IDT (6.7 μW) the range is 0.18–1.9. We therefore make no claim of a power advantage over EO driving. The remaining differences are qualitative: the acoustic resonator provides voltage gain without an external matching network, but at the cost of a ∼100-fold narrower RF bandwidth (Sec. IV) and a fixed shift frequency.

Comparison with acousto-optic shifters.—The best measured TFLN acousto-optic shifter reaches 3.5% on-chip efficiency at 30 dBm (1 W) [12]. The present design reaches, in simulation, $\eta_{\mathrm{abs}} = 55\%$ ($\eta_{\mathrm{shift}} = 98.3\%$) at 6.7 μW under ideal conditions and a median on-chip $\eta_{\mathrm{abs}} = 32.3\%$ at 33.7 μW ($Q_m = 2000$, 10% IDT efficiency; heater-on $\kappa_i \times 2$ Monte Carlo) under the realistic conditions of Sec. V. The comparison is not like for like: Ref. [12] is a broadband, non-resonant traveling-wave deflector with a 70-MHz microwave bandwidth, whereas our design is a simulated, doubly resonant (optical and acoustic) device with a ∼1-MHz RF bandwidth and a single fixed shift frequency; a measured device will also face the fabrication and locking requirements of Sec. V. Zhu *et al.* [16] operate an acoustic photonic molecule at milliwatt acoustic power in a backward, reflective geometry and do not report a forward shift efficiency or insertion loss. Table III summarizes these benchmarks.

TABLE I. Simulated ideal operating points (upper bounds; see Table IV and Sec. V for realistic conditions) at $\gamma/2\pi = 0.6$ GHz, $\kappa_i/2\pi = 80.75$ MHz (`data/results_B.json`) and the measured EO shifter of Ref. [9].

	AO (sim.)	EO[a] (sim.)	Hu [9] (meas.)
$\Omega_m/2\pi$ (GHz)	3.33	3.33	28.2
$\gamma/2\pi$ (GHz)	0.60	0.60	5.31
$\kappa_i/2\pi$ (MHz)	80.75	80.75	170
η_{shift}	98.3%	99.6%	98.7%
η_{abs}	55%	56%	—
IL (dB)	2.5	2.5	0.45
Carrier (dB)[g]	−23.2	−26.0	—
Largest spur (dB)[g]	−21.2	−30.6	—
P_{RF}, as-built feed	6.7 μW[b]	3.1 mW[c]	96 mW
P_{RF}, matched feed	3.4 μW[d]	1.2–13 μW[e]	—
RF bandwidth	∼1 MHz	≈122 MHz[f]	>3 GHz

[a] Push-pull, same optics. [b] Shao IDT (≈50% coupling). [c] $V_c = V_0$. [d] Ideal IDT. [e] LC match, ideal inductor to $Q_L = 30$. [f] LC match with $Q_L = 30$; the unmatched electrode is broadband. [g] Relative to the shifted line.

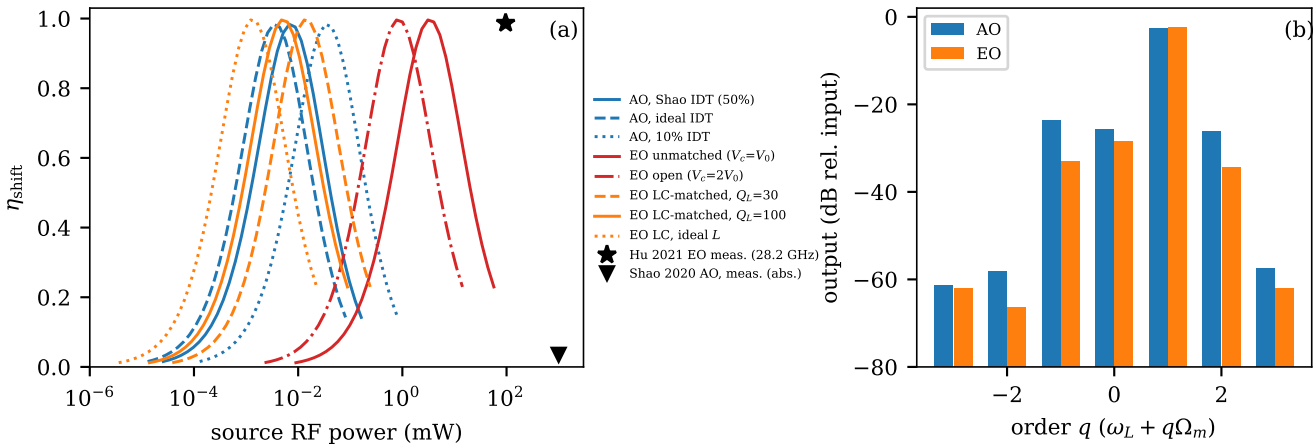


FIG. 2. Simulated operating point ($\gamma/2\pi = 0.6$ GHz, $\kappa_i/2\pi = 80.75$ MHz). (a) η_{shift} versus source RF power for the acoustic drive (blue: Shao IDT with ≈50% coupling, ideal IDT, 10% IDT) and for the EO drive with identical optics (red: unmatched, $V_c = V_0$, and open capacitor, $V_c = 2V_0$; orange: LC matched with inductor $Q_L = 30$, $Q_L = 100$, and an ideal inductor). Symbols mark measured devices: the EO shifter of Ref. [9] (star; 98.7% η_{shift} at 96 mW and 28.2 GHz). The Shao 2020 traveling-wave AO result (3.5% on-chip *absolute* efficiency at 1 W, 70 MHz bandwidth; this work is simulated at ∼1 MHz) is *not* placed on the η_{shift} axis; it is compared only under absolute efficiency. (b) Output power in each order q relative to the input at the respective optima of the AO and EO drives. The largest AO spur remains $q = -1$.

TABLE II. Source RF power at the optimum η_{shift} (simulation; `data/results_B.json`, `data/eff_vs_power.csv`).

Drive and feed	P_{RF}
AO, Shao IDT (≈50%)	6.7 μW
AO, ideal IDT (100%)	3.4 μW
AO, 10% IDT	34 μW
EO, unmatched ($V_c = V_0$)	3.1 mW
EO, open capacitor ($V_c = 2V_0$)	0.77 mW
EO, LC match, $Q_L = 30$	13 μW
EO, LC match, $Q_L = 100$	4.8 μW
EO, LC match, ideal inductor	1.2 μW
Hu *et al.* [9] (meas., 28.2 GHz)	96 mW

IV. TOLERANCES

Optical quality factor.—Because the coupling γ is fixed while κ_i varies, the shift ratio is robust but the loss is not [Fig. 3(a)]. For $Q_i \gtrsim 1.3 \times 10^6$ we find $\eta_{\mathrm{shift}} \approx 98\%$; at $Q_i = 6.4 \times 10^5$ it drops to 82%. The absolute efficiency rises from 33% at $Q_i = 1.3\times10^6$ through 55% at 2.4×10^6 to 87% at 9.7×10^6 (IL 0.56 dB). The required source power stays between 6.1 and 8.5 μW over this range.

RF detuning.—The acoustic resonance filters the drive. With $Q_m = 2000$ (FWHM 1.67 MHz) we find $\eta_{\mathrm{shift}} \geq 96.7\%$ within ±0.4 MHz of the doublet splitting, 80% at 1 MHz, 44% at 2 MHz, and 10% at 5 MHz [Fig. 3(b)]. The usable RF bandwidth is thus ∼1 MHz. Under the flat-drive assumption, the LC-matched EO drive keeps $\eta_{\mathrm{shift}} = 99.6\%$ over the full ±5 MHz range.

IDT efficiency.—The acoustic source power scales as $1/\eta_{\mathrm{IDT}}$ [Fig. 3(c)]: 3.4 μW at 100%, 6.7 μW at the Shao IDT's ≈50%, and 34 μW at 10%, a value reported for other LN acousto-optic microrings [20]. The acoustic drive needs more power than the $Q_L = 30$ LC-matched EO drive (13 μW) once $\eta_{\mathrm{IDT}} \lesssim 25\%$, more than the $Q_L = 100$ match (4.8 μW) below ≈70%, and more than the ideal-inductor match (1.2 μW) for any physical $\eta_{\mathrm{IDT}} \leq 100\%$.

Thermal drift and ring mismatch.—We extend Eq. (1) with a common laser–ring detuning Δ, a resonance mismatch δ_r of ring 2 relative to ring 1, and a splitting error $2\mu - \Omega_m$, keeping the drive at the operating point (Fig. 4); at zero offset we recover $\eta_{\mathrm{shift}} = 98.3\%$. We find $\eta_{\mathrm{shift}} \geq 95\%$ for $-72 \leq \Delta/2\pi \leq +55$ MHz ($\geq 90\%$ for -97 to $+85$ MHz), for $-104 \leq \delta_r/2\pi \leq +145$ MHz, and for $-96 \leq (2\mu - \Omega_m)/2\pi \leq +53$ MHz. Within the $\geq 95\%$ region the IL stays between 2.50 and 2.66 dB. To convert to temperature we use $|df/dT| \approx f\,(dn_e/dT)/n_g \approx 2.8$ GHz/K, with $f = 193.4$ THz and $dn_e/dT = 3.34 \times 10^{-5}$ K^{-1} at 1523 nm [21]. With the thermo-optic effect alone, a common drift must be held within ≈ ±20 mK (±30 mK for $\eta_{\mathrm{shift}} \geq 90\%$) and the differential ring temperature within ±37 mK. Thermal expansion, $\alpha_a \approx 1.54 \times 10^{-5}$ K^{-1} [22], shifts the resonances in the same direction; with $|df/dT| \approx f\,(dn_e/dT + n_{\mathrm{eff}}\alpha_a)/n_g \approx 5.6$ GHz/K ($n_{\mathrm{eff}} = 2.14$ assumed), consistent with the 5.6 GHz/K predicted for e-polarized modes of a bulk LN resonator [23], the windows shrink to ±9.8 mK (±15 mK for $\eta_{\mathrm{shift}} \geq 90\%$) for common drift and ±18.6 mK for the differential temperature. A bonded thin film is partly clamped by the substrate, so the bulk expansion makes these narrower windows con-

TABLE III. Benchmarks. "Meas." denotes measured devices, "sim." this work; LNOS, LN-on-sapphire; BW, RF bandwidth. All except Ref. [16] are on TFLN. The efficiencies are not all defined identically (see notes).

Work	Mechanism	Shift	Efficiency	Drive	Note
Shao *et al.* [12] (meas.)	AO traveling wave	∼3 GHz	3.5% on-chip[a]	1 W (30 dBm)	70 MHz BW
Zhu *et al.* [16] (meas.)	AO backward Brillouin (LNOS)	8.54 GHz	reflectivity ≤24%	mW (acoustic)	no forward η_{shift}
Hu *et al.* [9] (meas.)	EO photonic molecule	28.2 GHz	$\eta_{\text{shift}} = 98.7\%$	96 mW	IL 0.45 dB, >3 GHz BW
Qiu *et al.* [11] (meas.)	EO serrodyne	—	>98.8%	$V_\pi \approx 2.3$ V	broadband[b]
This work (sim.)[c]	AO photonic molecule	3.33 GHz	$\eta_{\text{shift}} = 96.4\%$, $\eta_{\text{abs}} = 32.3\%$	33.7 μW	IL 4.72 dB, ∼1 MHz BW
This work, ideal	upper bound	3.33 GHz	$\eta_{\text{shift}} = 98.3\%$, $\eta_{\text{abs}} = 55\%$	6.7 μW	IL 2.5 dB
EO ref. (sim.)	EO molecule, LC matched	3.33 GHz	$\eta_{\text{shift}} = 99.6\%$	1.2–13 μW	122 MHz BW ($Q_L = 30$)

[a]Deflected power divided by transmitted power with the microwave off [12]. [b]Values taken from the published abstract of Ref. [11]. [c]Heater-on, $\kappa_i \times 2$ Monte Carlo medians (Sec. V; v2).

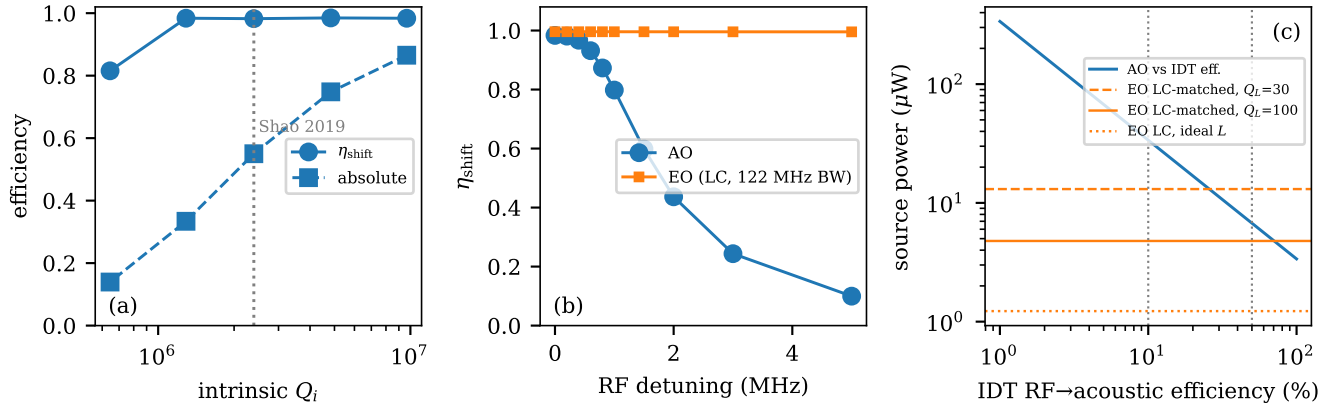


FIG. 3. Tolerances of the acoustic design (simulation). (a) η_{shift} (circles) and η_{abs} (squares) versus intrinsic Q_i, with $\delta\omega$ re-optimized at each point; the dotted line marks the Q_i implied by Ref. [13]. (b) η_{shift} versus RF detuning of the drive from the doublet splitting. For AO the drive amplitude follows the acoustic Lorentzian ($Q_m = 2000$, FWHM 1.67 MHz); the EO curve assumes a flat drive within the 122-MHz bandwidth of the $Q_L = 30$ LC match. (c) Source power of the acoustic drive versus IDT RF-to-acoustic efficiency, compared with the LC-matched EO drive (horizontal lines). Dotted vertical lines mark 10% and 50%.

servative; the true values lie between the two estimates. The mismatch is partly absorbed by retuning the laser along the ridge $\Delta \approx \delta_r/2$, and can be removed with a heater on ring 2. The splitting error, $\approx \pm 1.6\%$ of Ω_m, is set by the coupling gap and is thus a fabrication tolerance; the RF frequency can track the actual 2μ only within the IDT and acoustic-resonance bandwidth.

V. REALISTIC CONDITIONS: MONTE CARLO ANALYSIS

Each sample draws the ring–ring gap first, forms μ_{gap}, then (heater on) minimizes $|2\mu - \Omega_m|$ within $\Delta\phi \in [0, \pi]$ before lock jitter and the two IL columns. With $\mu_{\text{gap}} = \mu\sqrt{2}$ at the design gap, the heater hits the RF retuning window for 100% of the $N = 1000$ heater-on samples. Median heater power is ≈22 mW; crosstalk enters δ_r and Δ.

Heater on (main prediction).—Under IL column B and 10% IDT efficiency, the median $\eta_{\text{shift}} = 96.4\%$ (10th–90th percentile 91.7–98.3%); 94.0% of samples reach $\eta_{\text{shift}} \geq 90\%$ and 67.8% reach $\geq 95\%$. Median on-chip absolute efficiency is 32.3% with median IL 4.72 dB; fiber-to-fiber (1.5 dB/facet) is 16.2%. IL column A (Shao Q_i only) gives median IL 2.48 dB and on-chip absolute efficiency 55.0%. At $\xi = 1$, 10% IDT, $Q_m = 1570$, the source power is 43 μW; the ξ sweep at the same IDT and Q_m gives 478/172/43/11 μW for $\xi = 0.3/0.5/1/2$ (Fig. 6). The largest spurious sideband remains $q = -1$ in 100% of samples (Fig. 7). The heater crosstalk (2 mK differential / 3 mK common at P_π) remains an assumption pending calibration.

Heater off (comparison only).—With the coupler fixed, median $\eta_{\text{shift}} = 94.5\%$ (82.5–97.9%) and only 74.2% of samples at $\eta_{\text{shift}} \geq 90\%$. That case is not the fab prediction; it shows why a tunable coupler is required when the raw $2\mu - \Omega_m$ scatter is ∼100 MHz rms.

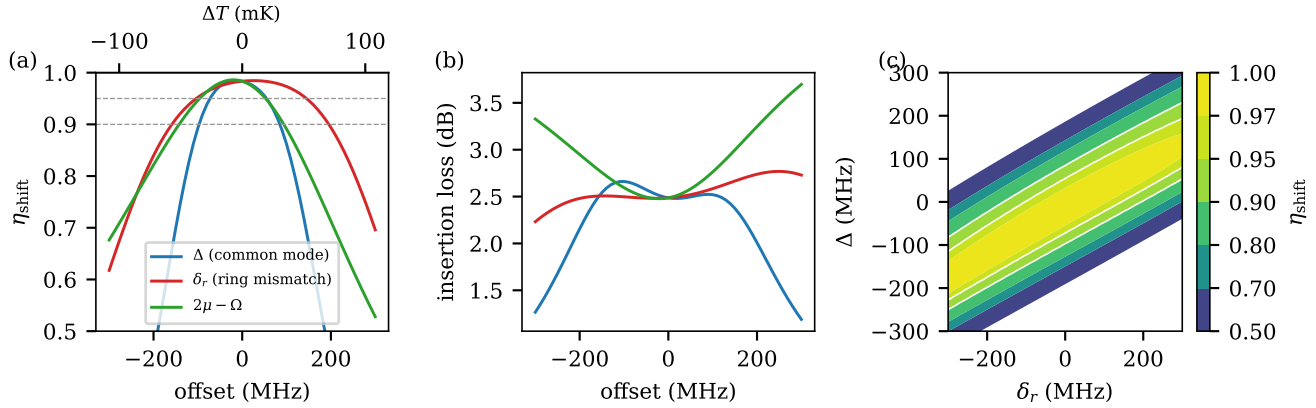


FIG. 4. Thermal-drift, ring-mismatch, and splitting tolerances of the acoustic design at the AO operating point (simulation; `data/tol_scan.csv`). η_{shift} versus the common detuning Δ of the laser from both rings, the ring-2–ring-1 resonance mismatch δ_r, and the splitting error $2\mu - \Omega_m$; the two-dimensional map shows the ridge along $\Delta \approx \delta_r/2$. Temperature axes use $|df/dT| \approx 2.8$ GHz/K (thermo-optic effect only); including thermal expansion (≈ 5.6 GHz/K) halves all temperature windows (see text).

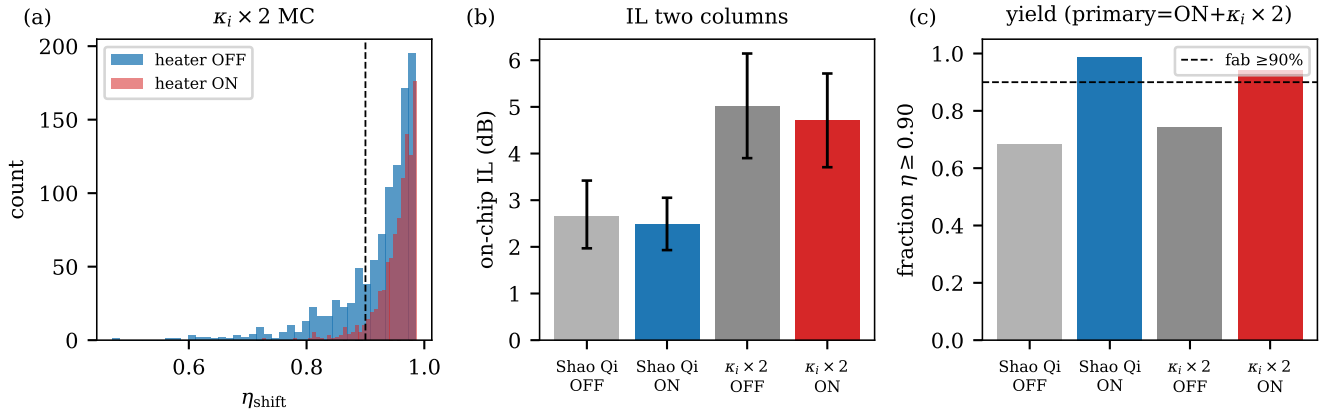


FIG. 5. Heater-on versus heater-off Monte Carlo (simulation, $N = 1000$; `data/summary_heater_xi.json`). Main prediction is heater on with conservative $\kappa_i \times 2$. Heater-off medians are comparison only. Shao 2020's 3.5% is an absolute efficiency and is not plotted on an η_{shift} axis.

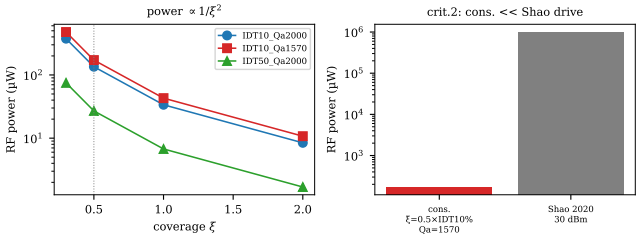


FIG. 6. Source RF power versus partial-coverage ξ at $\eta_{\text{IDT}} = 10\%$ and $Q_m = 1570$ (simulation; power $\propto 1/\xi^2$). Conservative lab margin: $\xi = 0.5$ gives ≈ 172 μW.

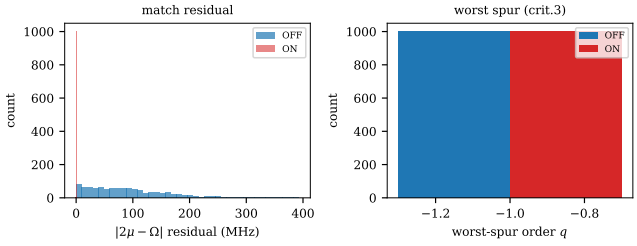


FIG. 7. Residual splitting after heater tuning and worst spurious order (simulation). Heater on: $|2\mu - \Omega_m|$ inside the RF window for all samples; worst spur is $q = -1$ throughout.

TABLE IV. Parameters of the heater/ξ Monte Carlo (`data/summary_heater_xi.json`). "Assumed" marks values without a direct measurement on this coupler geometry.

Parameter	Value	Source
IDT efficiency	10% (50% ref.)	Shao 2019 cal.; 10% conservative
Acoustic Q_m	2000 (3600 upper; 1570 cons.)	[13]; 1570 assumed
IL column A	Shao 2019 Q_i as measured	suspended racetrack with IDT
IL column B	extra $\kappa_i \times 2$	assumed (possible double count)
Partial coverage ξ	1 (sweep 0.3–2)	$\xi = 1$: Shao $V_\pi L$ extrapolation
MZI coupler	$\mu = \mu_{\rm gap} \sin(\Delta\phi/2)$	self-labeled; not [10]
Design $\mu_{\rm gap}$	$\mu\sqrt{2}$	assumed headroom
Coupler P_π	44.4 mW	Maeder geometric assumption [19]
Heater crosstalk	2 mK diff. / 3 mK CM at P_π	assumed (v2 isolation; calibrate)
Residual lock jitter	5 mK rms	assumed
Coupling-gap error σ	5 nm	assumed
RF retuning	$\pm$1 MHz	$\sim$half acoustic FWHM

VI. MEASUREMENT PLAN

Table V is the experimental interface for the tape-out. Predictions are from `data/summary_heater_xi.json` (v2 heater/ξ Monte Carlo; seed master 20261114) and related sims; the Measurement column is left blank. Stage-2 narrowband HOM is attempted only if step 4 matches prediction. Internal tape-out gates (yield $\geq$ 90% at $\eta_{\rm shift} \geq 90\%$; conservative RF $\ll$1 W Shao drive; spur $q = -1$) are checked in Notes under the *assumed* coupler-isolation crosstalk (2/3 mK at P_π) and are *not* claimed as abstract achievements or measured results: under that assumption all three gates pass (yield 94.0%; $\approx$172 μW; spur $q = -1$), but step 1 must calibrate heater power versus splitting/detuning before treating the crosstalk model as validated.

VII. STAGE-2 PREDICTIONS: HOM AND THERMAL PHONONS

HOM visibility (stage 2).—Narrowband HOM is a second-stage prediction, not a first-sample deliverable. A frequency shifter for quantum networks must preserve the spectral-temporal mode of the photon. We model an input photon with a Gaussian spectral amplitude $\psi(\Delta) \propto \exp(-\Delta^2/4\sigma^2)$, where σ is the rms width of $|\psi|^2$, and compute the complex transfer function $H(\Delta)$ of the shifted ($q = +1$) channel from the full time-domain solution of Eq. (1) at the AO operating point, evaluated on 41 points over $\pm 4\sigma$ (`data/H_shift.csv`). The shifted photon, filtered to its target frequency bin, interferes with a reference photon of the same spectrum at the target frequency. Because a fixed delay of the reference can always be added, we maximize over the delay τ:

$$V = \max_{\tau} \frac{\left| \int \psi^*(\Delta) H(\Delta) \psi(\Delta) e^{-2\pi i \Delta \tau} d\Delta \right|^2}{\int |\psi|^2 d\Delta \int |H\psi|^2 d\Delta}. \quad (3)$$

The normalization by $\int |H\psi|^2$ assumes that the loss is balanced, e.g. by attenuating the reference arm, so V measures mode distinguishability only. The optimum delay is 1.0–1.1 ns, i.e. the group delay of the device.

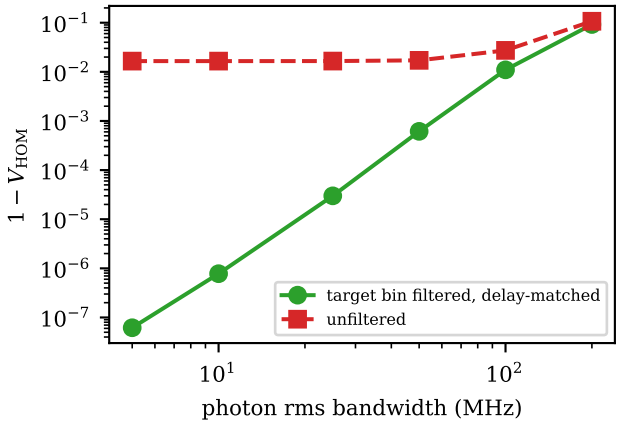


FIG. 8. HOM infidelity $1-V$ versus photon rms bandwidth σ (simulation; `data/HOM.csv`). Circles: shifted photon filtered to the target bin, reference photon delay-matched [Eq. (3)]. Squares: unfiltered, where the residual unshifted and spurious orders limit V to $\approx \eta_{\rm shift}$.

The results are shown in Fig. 8. With filtering, $1-V = 8\times10^{-7}$ at $\sigma = 10$ MHz, 3×10^{-5} at 25 MHz, 6×10^{-4} at 50 MHz, 1.1×10^{-2} at 100 MHz, and 9×10^{-2} at 200 MHz. The residual distinguishability comes from the curvature of $H(\Delta)$ beyond linear phase across the photon spectrum, which grows rapidly as σ becomes a sizable fraction of the optical linewidths ($\gamma/2\pi = 0.6$ GHz). Without filtering, the residual carrier and spurious orders limit the visibility to $V \approx \eta_{\rm shift}$: $V = 0.9835$ at $\sigma = 10$ MHz and 0.9829 at 50 MHz, i.e. $1 - V \approx 1.7 \times 10^{-2}$, more than four orders of magnitude worse than the filtered value at 10 MHz ($V = 0.9999992$). A spectral filter on the target bin is therefore required for high-visibility operation. The device is therefore suited to narrowband photons with $\sigma \lesssim 50$ MHz, such as cavity-enhanced parametric sources or atom- and quantum-dot-like emitters, together with a spectral filter on the target bin. Broadband photons require a wider optical linewidth, i.e. a larger γ and correspondingly larger drive.

Thermal phonons.—The shift is a pump-free, number-conserving, beam-splitter-type process: the acoustic field scatters a photon from the S to the AS supermode, and no photon is created without an input photon. Thermal occupation $n_{\rm th}$ of the acoustic mode therefore

TABLE V. Five-step measurement plan. Prediction = simulation; Measurement blank. Pass criteria gate the next step; they are not abstract claims of success.

Step	What to measure	Pass criteria	Prediction (sim.) / Measurement
1	Heater power vs. supermode splitting 2μ	Can pull 2μ into the acoustic FWHM ($\approx$1.7 MHz at $Q_m = 2000$)	Pred.: 100% of samples hit $\lvert 2\mu - \Omega\rvert$ window with $\mu_{\rm gap} = \mu\sqrt{2}$, $P_\pi = 44.4$ mW (Maeder geom. assump.); Meas.:
2	IDT S_{11}, acoustic Q	Q_m in 1570–3600; record $\eta_{\rm IDT}$ separately	Pred.: $Q_m = 2000$ typical, 3600 max [13]; Meas.:
3	RF voltage vs. optical sidebands; extract ξ	If $\xi \geq 0.5$, conservative power stays on the $1/\xi^2$ curve and the device remains measurable	Pred.: ξ = 0.3/0.5/1/2 $\rightarrow$ 478/172/43/11 μW (IDT 10%, $Q_m = 1570$); Meas.:
4	$\eta_{\rm shift}$, carrier, $q = -1$, on-chip IL	Use MC 10th percentile as lower accept limit; do not treat the median as guaranteed	Pred. (heater on, $\kappa_i \times 2$): $\eta_{\rm shift}$ p10= 91.7%, median 96.4%; IL median 4.72 dB (Shao Q_i: 2.48 dB); worst spur $q = -1$; Meas.:
5	Narrowband quantum HOM	Only if step 4 matches prediction; unfiltered V should be close to $\eta_{\rm shift}$	Pred.: unfiltered $1 - V \approx 1 - \eta_{\rm shift}$; filtered $1 - V = 8 \times 10^{-7}$ at 10 MHz (Sec. VII); Meas.:

adds an incoherent fluctuation of the coupling of relative size $n_{\rm th}/N_{\rm coh}$, where $N_{\rm coh} = (4\gamma_e/\gamma^2)\, P_{\rm RF}/\hbar\Omega_m$ is the coherently driven phonon number. With $\gamma = \Omega_m/Q_m$, $Q_m = 2000$, and $\gamma_e/\gamma = 0.15$ [13], we find $N_{\rm coh} \approx 1.75 \times 10^{11}$ at the operating point. At 3.33 GHz, $n_{\rm th} = 1877$ at 300 K, 24.5 at 4 K, and 1.1×10^{-7} at 10 mK, so $n_{\rm th}/N_{\rm coh} = 1.1 \times 10^{-8}$, 1.4×10^{-10}, and 7×10^{-19}, respectively. As a second term we bound spontaneous scattering mediated by the *same* acoustic mode by the single-photon cooperativity $C_0 = 4g_0^2/(\kappa\gamma_m)$, with $g_0/2\pi = 1.1$ kHz [13], $\kappa/2\pi = (\gamma + \kappa_i)/2\pi = 680.75$ MHz, and $\gamma_m = \Omega_m/Q_m$ ($\gamma_m/2\pi = 1.665$ MHz), giving $C_0 = 4.27 \times 10^{-9}$ and $n_{\rm th}C_0 = 8.0 \times 10^{-6}$ at 300 K (`data/tapeout_summary.json`). This is still $\sim 2000\times$ smaller than the unfiltered sideband error $1 - \eta_{\rm shift} = 0.0165$ at the ideal point, so neither term dominates the residual carrier/spur budget. Scattering into *other* optical modes is a different channel and is not claimed to be covered by $n_{\rm th}C_0$; that multimode estimate remains open. Unlike microwave-to-optical transduction, thermal phonons are not converted into noise photons by the beam-splitter interaction itself.

VIII. LIMITATIONS

The results above are design predictions subject to the following limitations.

(i) *No power advantage over matched EO.* With impedance matching on both sides the acoustic and EO source powers differ only by a factor of 0.36–3.9, and the LC-matched EO drive is more power-efficient with a low-loss inductor (Sec. III). The advantage claimed here is restricted to the class of acousto-optic frequency shifters.

(ii) *Narrow RF bandwidth and fixed shift frequency.* The usable RF bandwidth is $\sim$1 MHz, set by the acoustic $Q_m = 2000$, versus $\approx$122 MHz for a $Q_L = 30$ LC-matched EO electrode and >3 GHz for the unmatched device of Ref. [9]. The shift is fixed at the 3.33-GHz acoustic resonance, whereas EO photonic molecules have been operated at 12.5–28.2 GHz [9]. The RF source must be frequency-stabilized, and the doublet splitting must be thermally tuned to match Ω_m to well within the optical linewidth. Maintaining $\eta_{\rm shift} \geq 95\%$ requires active stabilization of the common resonance at the $\pm$9.8 mK level when thermal expansion is included ($\pm$20 mK for the thermo-optic effect alone; Sec. IV), e.g. with heater electrodes or by locking the laser to the resonance; passive temperature control is insufficient. The ring–ring mismatch, in contrast, can be corrected with a heater and is not a fundamental limit.

(iii) *Insertion loss.* The ideal IL is 2.5 dB ($\eta_{\rm abs} = 55\%$), limited by the measured $Q_i \approx 2.4 \times 10^6$ of the suspended platform [13]; under realistic heater-on $\kappa_i \times 2$ conditions the median is 4.72 dB, compared with 0.45 dB for the measured EO device [9]. The fiber-to-fiber estimate assumes 1.5 dB per facet, which requires an optimized edge coupler [24]; the suspended chip of Ref. [13] had $\lesssim$5 dB per facet.

(iv) *IDT efficiency.* The acoustic drive loses to a $Q_L = 30$ LC-matched EO drive if the IDT efficiency is below $\approx$25%; IDT efficiencies of $\approx$10% have been reported in related LN microring devices [20].

(v) *Drive extrapolation and ξ.* We obtain $\delta\omega$ by extrapolating the measured MZI $V_\pi L$ of Ref. [13] linearly to a ring that the acoustic field covers only partially (15%), encoded as $\xi = 1$. Smaller ξ raises the required source power as $1/\xi^2$; step 3 of Table V extracts ξ from RF voltage versus optical sidebands. We assume the same IDT efficiency, mode overlap, and uniform acoustic phase along the covered section as in the MZI; $n_g = 2.3$ is assumed.

(vi) *Heater residual and lock jitter (Sec. V).* With the MZI coupler heater on and $\mu_{\rm gap} = \mu\sqrt{2}$, the Monte Carlo predicts that $\lvert 2\mu - \Omega_m\rvert$ falls inside the $\pm$1 MHz RF window for all samples at 5 nm gap σ; the heater-off

comparison retains ∼100 MHz rms splitting error (median $\eta_{\text{shift}} = 94.5\%$, 74.2% at $\eta_{\text{shift}} \geq 90\%$). Residual risks are the *assumed* heater crosstalk into δ_r and Δ (2 mK differential / 3 mK common at P_π after a coupler-isolation redesign; v1 used 5/10 mK), the 5 mK lock jitter, and whether the fabricated coupler provides the assumed headroom. Under conservative $\kappa_i \times 2$ the heater-on yield at $\eta_{\text{shift}} \geq 90\%$ is 94.0% (above the internal 90% fab gate) together with the power and spur gates, so simulation supports proceeding to layout with measurement hooks—provided step 1 of Table V calibrates heater power versus splitting/detuning and does not treat the crosstalk numbers as measured. These are predictions for steps 1 and 4, not a stop at "5 nm, unmodeled."

(vii) *Assumed values.* The extra optical loss, residual lock jitter, gap error, coupling decay length, Q_i spread, and the conservative acoustic $Q_m = 1570$ (no literature source) are assumptions without direct measurements on this platform (Table IV); we report sensitivity scans for each. The extra-loss term is possibly double counted, because the measured Q_i of Ref. [13] already comes from a suspended device with an IDT, so the realistic IL is likely pessimistic. The bulk thermal-expansion coefficient likely overestimates the expansion of a bonded film.

(viii) *Not modeled.* We did not model ohmic and acoustic heating of the IDT and resonator, the resulting drive-dependent drift of the ring resonances beyond the lock jitter above (Ref. [13] observed a heating-induced red shift with increasing microwave power), or the mechanical stability of the suspended structure.

(ix) *Model provenance.* Our implementation of Eq. (1) is reconstructed from the published equations of Ref. [9], not from the authors' code, and the design combines parameters of two different devices (optics and electrodes from Ref. [9], acoustics and loss from Ref. [13]). Any extension that routes the acoustic drive through on-chip phononic circuits (for example, to feed several shifters) would at present rely on acoustic-circuit parameters borrowed from GaN-on-sapphire [25] and on our reconstruction of that model from its equations; none of the numbers reported here depend on it.

(x) *Simulation only.* No device has been fabricated; every number in this paper, including the realistic Monte Carlo, is a model prediction.

IX. CONCLUSION

We have given a fabrication-oriented design for a suspended-IDT acoustic resonator driving one ring of a TFLN photonic molecule as a forward, single-bus frequency shifter at 3.33 GHz, together with tolerances and a five-step measurement plan. All numbers are predictions. Relative to an EO drive of the same optics, the Shao IDT needs ≈6.7 μW versus 1.2–13 μW when the EO feed is matched: there is no claimed AO power advantage, and the AO bandwidth is ∼1 MHz. Relative to Shao 2020, we compare absolute efficiency only (simulation, ∼1 MHz versus measured 3.5% at 1 W with 70 MHz). With the heater-on self-labeled MZI coupler, conservative loss, and 10% IDT, the Monte Carlo median η_{shift} is 96.4% (p10 91.7%); heater-off 94.5% is a comparison only. IL is quoted both with Shao 2019 Q_i (median 2.48 dB) and with assumed extra $\kappa_i \times 2$ (median 4.72 dB). Conservative RF power at $\xi = 0.5$, $Q_m = 1570$, and 10% IDT is ≈172 μW. Unfiltered stage-1 spectroscopy should track $1 - V \approx 1 - \eta_{\text{shift}}$; filtered HOM ($1 - V = 8 \times 10^{-7}$ at 10 MHz) is stage 2. Thermal phonons contribute $n_{\text{th}}/N_{\text{coh}} \approx 1.07 \times 10^{-8}$ and $n_{\text{th}} C_0 \approx 8.0 \times 10^{-6}$. Unlike the backward-Brillouin photonic molecule of Ref. [16], the present design is a forward, single-bus shifter and reports η_{shift} explicitly. The five measurement steps of Table V—heater tuning of 2μ, IDT S_{11}/Q_m, extraction of ξ, η_{shift}/IL/spurs, then narrowband HOM—are the path from this prediction to data. Under the assumed coupler-isolation crosstalk, an internal tape-out check on the conservative-loss Monte Carlo passes all three gates (yield 94.0% at $\eta_{\text{shift}} \geq 90\%$; ≈172 μW; spur $q = -1$): keep the AO device concept and proceed to layout with those measurement hooks, but calibrate the heater crosstalk before treating the yield gate as validated.

Data availability.—Simulation scripts (`core.py`, `make_all.py`, `hom_post.py`) and data under `data/` (including `summary_heater_xi.json`) are available from the author.

[1] J. M. Lukens and P. Lougovski, Frequency-encoded photonic qubits for scalable quantum information processing, Optica **4**, 8 (2017).

[2] M. Kues, C. Reimer, P. Roztocki, L. R. Cortés, S. Sciara, B. Wetzel, Y. Zhang, A. Cino, S. T. Chu, B. E. Little, D. J. Moss, L. Caspani, J. Azaña, and R. Morandotti, On-chip generation of high-dimensional entangled quantum states and their coherent control, Nature **546**, 622 (2017).

[3] H.-H. Lu, J. M. Lukens, N. A. Peters, O. D. Odele, D. E. Leaird, A. M. Weiner, and P. Lougovski, Electro-optic frequency beam splitters and tritters for high-fidelity photonic quantum information processing, Phys. Rev. Lett. **120**, 030502 (2018).

[4] C. Wang, M. Zhang, X. Chen, M. Bertrand, A. Shams-Ansari, S. Chandrasekhar, P. Winzer, and M. Lončar, Integrated lithium niobate electro-optic modulators operating at CMOS-compatible voltages, Nature **562**, 101 (2018).

[5] M. Zhang, C. Wang, R. Cheng, A. Shams-Ansari, and M. Lončar, Monolithic ultra-high-Q lithium niobate microring resonator, Optica **4**, 1536 (2017).

[6] D. Zhu, C. Chen, M. Yu, L. Shao, Y. Hu, C. J. Xin, M. Yeh, S. Ghosh, L. He, C. Reimer, N. Sinclair, F. N. C. Wong, M. Zhang, and M. Lončar, Spectral control of nonclassical light pulses using an integrated thin-film lithium niobate modulator, Light Sci. Appl. **11**, 327 (2022).

[7] R. Yang, W. Zhou, D.-J. Guo, H.-M. Ke, L. Tao, Y. Wei, *et al.*, Quantum photonic frequency processor on thin-film lithium niobate (2026), arXiv:2603.11471.

[8] L. M. Cohen, M. H. Muñoz-Arias, M. Sarovar, N. A. Boynton, S. C. Arterburn, T. A. Friedmann, N. T. Otterstrom, P. S. Davids, and M. Gehl, A heterogeneously integrated coupled-cavity frequency beam splitter (2026), arXiv:2608.26028.

[9] Y. Hu, M. Yu, D. Zhu, N. Sinclair, A. Shams-Ansari, L. Shao, J. Holzgrafe, E. Puma, M. Zhang, and M. Lončar, On-chip electro-optic frequency shifters and beam splitters, Nature **599**, 587 (2021).

[10] M. Zhang, C. Wang, Y. Hu, A. Shams-Ansari, T. Ren, S. Fan, and M. Lončar, Electronically programmable photonic molecule, Nat. Photonics **13**, 36 (2019).

[11] Z. Qiu, Z. Zhou, H. Zhong, L. Zhao, H. Zhang, L. Hu, G. Wu, and J. Chen, On-chip bidirectional optical frequency shifter for optical frequency transfer, Opt. Lett. **50**, 1132 (2025).

[12] L. Shao, N. Sinclair, J. Leatham, Y. Hu, M. Yu, T. Turpin, D. Crowe, and M. Lončar, Integrated microwave acousto-optic frequency shifter on thin-film lithium niobate, Opt. Express **28**, 23728 (2020).

[13] L. Shao, M. Yu, S. Maity, N. Sinclair, L. Zheng, C. Chia, A. Shams-Ansari, C. Wang, M. Zhang, K. Lai, and M. Lončar, Microwave-to-optical conversion using lithium niobate thin-film acoustic resonators, Optica **6**, 1498 (2019).

[14] S. A. Tadesse and M. Li, Sub-optical wavelength acoustic wave modulation of integrated photonic resonators at microwave frequencies, Nat. Commun. **5**, 5402 (2014).

[15] S. Kapfinger, T. Reichert, S. Lichtmannecker, K. Müller, J. J. Finley, A. Wixforth, M. Kaniber, and H. J. Krenner, Dynamic acousto-optic control of a strongly coupled photonic molecule, Nat. Commun. **6**, 8540 (2015).

[16] Z.-X. Zhu, Y.-H. Yang, X.-B. Xu, J.-Q. Wang, Y. Zeng, J.-H. Zou, J. Lu, W. Wang, M. Li, Y.-L. Zhang, G.-C. Guo, L. Sun, and C.-L. Zou, Acoustically control of integrated optical microrings: from photonic molecule to Mobius strip (2025), arXiv:2511.22585 [physics.optics].

[17] A. J. Shepherd and R. O. Behunin, Frequency-entangled W states and quantum frequency translation protocols via forward Brillouin interactions, Phys. Rev. A **111**, 042407 (2025).

[18] In backward SBS the written phonon grating extends over the signal–pump collision length, $L_c \approx 0.26$ m for 3-ns signal and 1-ns pump pulses, not over an acoustic wavelength. For a 10-GHz bin spacing the phase mismatch between bins is $\Delta q \approx 964$ rad/m, so $\Delta q\, L_c \approx 251$ and different bins write mutually orthogonal gratings. In our one-dimensional space–time simulation of a two-bin, mode-selective write, the selectivity between the targeted and the orthogonal bin superposition drops to 0.5, i.e. no selectivity. The scheme would also need meter-scale interaction lengths (≈1.3 m for 100-MHz photons) and watt-level pump power ($P_\pi \approx 1.55$ W at 100 MHz). These are our estimates from TFLN literature parameters and have not been tested experimentally.

[19] A. Maeder, G. Finco, F. Kaufmann, A. Sabatti, J. Kellner, R. J. Chapman, and R. Grange, On-chip tunable quantum interference in a lithium-niobate-on-insulator photonic circuit, Quantum Sci. Technol. **9**, 035040 (2024), $p_\pi = 44.4$ mW for ≈1 mm Au heater arm; geometric reference only, not this coupler layout.

[20] X.-B. Xu, Z.-X. Zhu, Y.-H. Yang, J.-Q. Wang, Y. Zeng, J.-H. Zou, J. Lu, Y.-L. Zhang, W. Wang, G.-C. Guo, L. Sun, and C.-L. Zou, Magnetic-free optical mode degeneracy lifting in lithium niobate microring resonators (2025), arXiv:2509.01940.

[21] L. Moretti, M. Iodice, F. G. Della Corte, and I. Rendina, Temperature dependence of the thermo-optic coefficient of lithium niobate, from 300 to 515 K in the visible and infrared regions, J. Appl. Phys. **98**, 036101 (2005).

[22] Y. S. Kim and R. T. Smith, Thermal expansion of lithium tantalate and lithium niobate single crystals, J. Appl. Phys. **40**, 4637 (1969).

[23] W. Weng, P. S. Light, and A. N. Luiten, Ultra-sensitive lithium niobate thermometer based on a dual-resonant whispering-gallery-mode cavity, Opt. Lett. **43**, 1415 (2018).

[24] D. Jia, Q. Luo, C. Yang, R. Ma, X. Yu, F. Gao, Q. Yang, F. Bo, G. Zhang, and J. Xu, High-efficiency edge couplers enabled by vertically tapering on lithium-niobate photonic chips, Appl. Phys. Lett. **123**, 263502 (2023).

[25] X.-B. Xu, Y. Zeng, J.-Q. Wang, Z.-H. Tian, J.-Z. Zhang, Y.-H. Yang, Z.-X. Zhu, J.-H. Zou, L. Xiao, W. Wang, B.-Z. Wang, G.-C. Guo, L. Sun, and C.-L. Zou, Large-scale programmable phononic integrated circuits (2025), arXiv:2510.26596.